

JAVA PRACTICAL RECORD

Experiment No.	05
Date	23/09/2026
Program	ConstructorOverloading
Concept	Constructor Overloading

1. Objective / Aim

To design and implement a Java program demonstrating constructor overloading by creating multiple constructors with different parameter lists in the Employee class.

2. System Requirements

Hardware: Computer/Laptop with basic processing capability.

Software: Java Development Kit (JDK) 8 or above and any Java IDE or command-line terminal.

3. Algorithm / Logic Description

1. Start the program.
2. Create an **Employee** class with name, id, designation, salary, and promotionStatus fields.
3. Define a no-argument constructor to initialize default employee details.
4. Define a constructor with name and ID parameters.
5. Define a constructor with name, ID, designation, and salary parameters.
6. Define a constructor with all five employee parameters.
7. Create four Employee objects using the four different overloaded constructors.
8. Call the **display()** method for each object.
9. Display the employee details.
10. Stop the program.

4. UML Class Diagram

Employee	ConstructorOverloading
- name : String - id : int - designation : String - salary : double - promotionStatus : String	- e1 : Employee - e2 : Employee - e3 : Employee - e4 : Employee
+ Employee() : Employee + Employee(name : String, id : int) : Employee + Employee(name : String, id : int, designation : String, salary : double) : Employee + Employee(name : String, id : int, designation : String, salary : double, promotionStatus : String) : Employee + display() : void	+ main(args : String[]) : void

Constructor overloading is demonstrated by defining four Employee constructors with different parameter lists. The main class creates four objects, each using a different constructor.

5. Source Code

```
class Employee {
    String name;
    int id;
    String designation;
    double salary;
    String promotionStatus;

    // Constructor 1: No arguments
    Employee() {
        name = "Unknown";
        id = 0;
        designation = "Not Assigned";
        salary = 0.0;
        promotionStatus = "Not Available";
    }

    // Constructor 2: Name and ID
    Employee(String name, int id) {
        this.name = name;
        this.id = id;
        designation = "Not Assigned";
        salary = 0.0;
        promotionStatus = "Not Available";
    }

    // Constructor 3: Name, ID, Designation and Salary
    Employee(String name, int id, String designation, double salary) {
        this.name = name;
        this.id = id;
        this.designation = designation;
        this.salary = salary;
        promotionStatus = "Pending";
    }

    // Constructor 4: All parameters
    Employee(String name, int id, String designation,
        double salary, String promotionStatus) {
        this.name = name;
        this.id = id;
        this.designation = designation;
        this.salary = salary;
        this.promotionStatus = promotionStatus;
    }

    void display() {
        System.out.println("Name           : " + name);
        System.out.println("ID           : " + id);
        System.out.println("Designation      : " + designation);
        System.out.println("Salary          : " + salary);
        System.out.println("Promotion Status : " + promotionStatus);
        System.out.println("-----");
    }
}

public class ConstructorOverloading {
    public static void main(String[] args) {

        // Calling different overloaded constructors
        Employee e1 = new Employee();

        Employee e2 = new Employee("Ravi", 101);

        Employee e3 = new Employee(
            "Sita", 102, "Manager", 60000
        );

        Employee e4 = new Employee(
            "Kiran", 103, "Senior Developer", 75000, "Promoted"
        );

        // Display employee details
        e1.display();
        e2.display();
        e3.display();
        e4.display();
    }
}
```

6. Compilation & Execution Commands

```
javac ConstructorOverloading.java
java ConstructorOverloading
```

7. Output Screen

```
Name          : Unknown
ID             : 0
Designation    : Not Assigned
Salary        : 0.0
Promotion Status : Not Available
-----
Name          : Ravi
ID            : 101
Designation    : Not Assigned
Salary        : 0.0
Promotion Status : Not Available
-----
Name          : Sita
ID            : 102
Designation    : Manager
Salary        : 60000.0
Promotion Status : Pending
-----
Name          : Kiran
ID            : 103
Designation    : Senior Developer
Salary        : 75000.0
Promotion Status : Promoted
-----
```

8. Result

The Java program was successfully implemented and executed. Constructor overloading was demonstrated by creating Employee objects using four constructors with different numbers and types of parameters.